Smart Restaurant Billing System

Project Statement | Python Console Application

1. Introduction

Billing is one of the most frequent and error-prone tasks in a restaurant. Adding up items by hand, working out discounts and calculating tax for every customer takes time and invites mistakes, especially during busy hours. The **Smart Restaurant Billing System** is a command-line application written in pure Python that automates this entire process.

The program is built around a fictional restaurant, *The Python Kitchen*. It shows a numbered menu, lets the cashier add items and quantities to an order, applies the right discount for the customer, adds tax and produces a clear, itemized receipt that is printed on screen and saved to a text file. It uses only Python's standard library, so it runs anywhere Python is installed without any extra setup.

2. Objectives

- Replace manual bill calculation with a fast, accurate and repeatable process.
- Apply discounts and tax consistently using clearly defined rules.
- Generate a neat receipt that is shown to the customer and saved for records.
- Serve as a beginner-friendly learning project that demonstrates core Python concepts.

3. Key Features

Menu display	Shows 12 numbered items (starters, biryanis, breads, drinks and desserts) with prices in Rs.
Order taking	The cashier enters an item number and quantity, repeating until finished by typing 0.
Input validation	Invalid numbers, unknown menu items and wrong yes/no answers are rejected with a prompt to try again.
Order summary	Lists everything ordered and asks for confirmation before the bill is generated.
Customer types	Regular (0%), Student (5%), Premium (15%) and Couple (10%) discounts.
Bulk discount	10% off when the subtotal is Rs.1000 or more.
Smart discount rule	The customer receives whichever discount is higher; the two are never stacked.
Tax	5% tax calculated on the amount after the discount.

Receipt	Formatted receipt with date and time, printed on screen and saved to receipt.txt.

4. How It Works

The user starts at a main menu with three options: place a new order, view the menu or exit. When a new order is placed, the program follows these steps:

- **Take the order:** items are chosen by number and stored as a list of dictionaries holding name, price and quantity.
- **Review:** an order summary is shown and the cashier confirms that the bill should be generated.
- **Choose customer type:** the cashier selects Regular, Student, Premium or Couple.
- **Calculate:** subtotal, then the larger of the bulk and customer discount, then tax on the discounted amount, then the final total.
- **Print and save:** the receipt text is built once, displayed and written to a file.

Every calculation lives in its own small function, and all settings such as the restaurant name, tax rate and discount rates are kept as constants at the top of the file, so the rules can be changed easily without touching the logic.

5. Technologies and Concepts Used

- **Language:** Python 3 (standard library only, using the datetime module).
- **Concepts:** dictionaries, lists, tuples, functions, while loops, conditionals, f-string formatting, input validation and file handling.
- **Design:** modular functions, constants for configuration, and the `if __name__ == "__main__"` entry-point pattern.

6. Conclusion

The Smart Restaurant Billing System shows how a small, well-organized Python program can solve a real everyday problem. It turns a slow manual task into a quick and dependable one, always applying discounts and tax correctly and leaving a saved receipt for every order. Because the code is modular and heavily commented, it is easy to read, test and extend.

Possible future improvements include keeping a separate receipt file for each bill, storing the menu in a file or database, adding a sales history and daily report, supporting table numbers, and building a graphical or web interface. As a foundation, the project gives students and beginners a practical example of how core programming concepts come together in a complete working application.